

ENFORCER

boat: arrow 2

A A fine becomes a price

Breaks where the rule meets behavior

MECHANISM

Fine

a charge for a violation

FAILURE

Paid, not obeyed

agents treat the fine as a fee for the act

MEASURE

contribution stability

CHECK

Does cooperation hold without the fine?

yes: keep and retest · no: add safeguards

SAFEGUARDS

- ✓ state the norm, not only the price
- ✓ graduated sanctions

MOST EXPOSED

incentive

↻ revise, then test again

OBSERVER

boat: arrow 4

B A score becomes the target

Breaks where behavior becomes a signal

MECHANISM

Reputation score

past conduct sets trust

FAILURE

Score is gamed

reputation farming; shallow cooperation

MEASURE

gameability; false negatives

CHECK

Does the score match real conduct?

yes: keep and retest · no: add safeguards

SAFEGUARDS

- ✓ tamper-evident records
- ✓ peer review
- ✓ audit messages, not only scores

MOST EXPOSED

social

epistemic

↻ revise, then test again

ARBITER

boat: the feedback loop

C A role becomes an authority

Breaks where the institution feeds back on itself

MECHANISM

Monitor or judge

a role that can sanction

FAILURE

Over-detects or is captured

rewarded for finding violations

MEASURE

false punishment; capture; override

CHECK

Are rulings accurate and reversible?

yes: keep and retest · no: add safeguards

SAFEGUARDS

- ✓ checks on sanctioners
- ✓ appeals
- ✓ elected or rotating roles
- ✓ a human stop

MOST EXPOSED

epistemic

constraint

↻ revise, then test again